

Application Layer

React + Three.js + Chart.js
~14K LOC TypeScript, client-side SPA



Simulation Layer

Spatial ABM (4 cell types)
10-factor reaction-diffusion field
Checkpoint-aware phagocytosis



AI Layer

Neural surrogate MLP (6-32-32-3)
Polarization map: context -> M1/M2
~9x speedup vs ODE (RK4)



Data Layer

scRNA-seq / TAM atlas / literature priors
Calibratable parameters, deterministic seed